

DPSeg: Dual-Prompt Cost Volume Learning for Open-Vocabulary Semantic Segmentation

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Abstract

Open-vocabulary semantic segmentation aims to segment images into distinct semantic regions for both seen and unseen categories at the pixel level. Current methods utilize text embeddings from pre-trained vision-language models like CLIP but struggle with the inherent domain gap between image and text embeddings, even after extensive alignment during training. Additionally, relying solely on deep text-aligned features limits shallow-level feature guidance, which is crucial for detecting small objects and fine details, ultimately reducing segmentation accuracy. To address these limitations, we propose a dual prompting framework, DPSeg, for this task. Our approach combines dual-prompt cost volume generation, a cost volume-guided decoder, and a semantic-guided prompt refinement strategy that leverages our dual prompting scheme to mitigate alignment issues in visual prompt generation. By incorporating visual embeddings from a visual prompt encoder, our approach reduces the domain gap between text and image embeddings while providing multi-level guidance through shallow features. Extensive experiments demonstrate that our method significantly outperforms existing state-of-the-art approaches on multiple public datasets. The code and models are available: <https://github.com/ziyuz-vision/DPSeg>.

1. Introduction

Semantic segmentation, the process of assigning class labels to individual pixels in an image, has seen significant advancements. Conventional segmentation methods [4, 5, 21, 25, 28, 42, 64, 65] are often structured for closed-set tasks, wherein models are trained and evaluated on a fixed set of categories. While these models demonstrate strong performance in controlled environments, they typically struggle to generalize effectively to real-world scenarios

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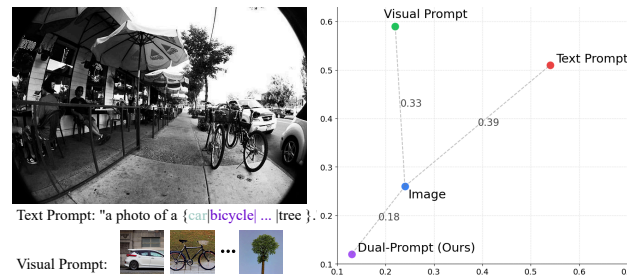


Figure 1. T-SNE visualization of embeddings in CLIP feature space, showing text prompt, visual prompt, target image, and our dual-prompt for a street scene image. Numbers indicate distance to image feature, where our dual-prompt approach achieves closest proximity (0.18).

ios that include unseen objects and novel categories.

Open-vocabulary semantic segmentation (OVSS) has emerged as a promising solution to this limitation. By leveraging large-scale vision-language models like CLIP [37], OVSS enables segmentation beyond fixed categories, allowing models to generalize to arbitrary classes during inference. Recent approaches, such as CAT-Seg, [7] utilize pixel-level cost volumes—dense similarity mappings between local image features and text embeddings in CLIP’s space—to balance segmentation accuracy and efficiency. SED [52] further enhances this framework by incorporating a hierarchical encoder and multi-scale features. However, existing methods face two key challenges: (1) despite incorporating multi-scale semantic information, the reliance on upsampling the cost volume leads to potential degradation of fine-grained details, particularly affecting small but crucial semantic segments, and (2) the sole dependence on text-image embedding alignment fails to fully resolve the domain gap, even with robust pre-trained models.

To overcome these limitations, our work begins by conducting comprehensive experiments (see Sec. 3.1 and Sec. 3.2) to investigate the domain gap between image and text embeddings by calculating cosine similarity scores across sampled categories. Our findings reveal that visual prompts exhibit stronger feature alignment with image

representations in the embedding space compared to text prompts. Additionally, the cost volumes generated from visual prompts demonstrate richer spatial-semantic cues for precise segment localization than their text-based counterparts. Building on these insights, we propose a novel OVSS framework with two key components: (1) a dual-prompt cost volume generation mechanism that leverages both visual and text prompts, and (2) a cost volume-guided decoder that progressively integrates visual prompt features at corresponding feature levels for refined segmentation prediction. To further improve segmentation accuracy and mitigate potential misalignment between visual prompts and actual segments during inference, we propose a semantic-guided prompt refinement strategy that employs a two-stage process, where initial segmentation results serve as refined visual prompts to guide the final prediction. As visualized in Fig. 1, our dual-prompt scheme achieves superior embedding space alignment with a distance of 0.18 to image features, compared to 0.33 for visual prompts and 0.39 for text prompts alone. The t-SNE visualization demonstrates that our dual-prompt approach effectively bridges the semantic gap by integrating synergistic information from both modalities. Extensive experiments show that our method sets a new benchmark in open-vocabulary semantic segmentation, achieving superior accuracy, inference efficiency, and overall performance. Our main contribution is the following:

- Through both quantitative and qualitative analysis of CLIP embeddings, we demonstrate that visual prompts achieve stronger feature alignment and generate more precise spatial-semantic cost volumes compared to text-based approaches.
- We propose **DPSeg**, a novel approach for open-vocabulary semantic segmentation featuring dual-prompt cost volume generation and cost volume-guided decoder. Additionally, our semantic-guided prompt refinement module leverages initial predictions as scene-specific visual prompts to enhance segmentation quality through a two-stage inference process.
- The extensive experiment results on multiple open vocabulary semantic segmentation datasets demonstrate that our method outperforms competitors by a large margin.

2. Related Work

2.1. Cross-Modality Learning Models

Cross-modality representation learning has seen significant advancements, broadly categorized into Spatial Cross Modality (2D-3D) [65] and Conceptual Cross Modality (Vision-Language). Vision-language models [6, 29, 39, 43] have been developed and fine-tuned on various downstream tasks using image-text pairs. CLIP [37, 58], for instance, leverages large-scale image-text data to align visual and language representations, excelling in zero-shot im-

age recognition and expanding into tasks like segmentation [15, 23, 54, 56], captioning [30], classification [1], restoration [60], and object detection [13, 66]. Recent CLIP-based extensions have shown notable improvements across various tasks. SLIP [32] enhances CLIP by incorporating self-supervised image-to-image contrastive learning alongside CLIP’s image-text alignment, yielding richer and more robust visual representations that capture subtle image features more effectively. A-CLIP [57] introduces an attentive token removal strategy, retaining only a small subset of tokens that exhibit strong semantic correlations with the corresponding text descriptions, allowing for more focused and relevant visual-text alignment.

In segmentation, OVSeg [23] adapts CLIP by fine-tuning on masked image regions to produce mask-aware embeddings for object-level segmentation. However, it struggles to capture detailed semantic information, such as object attributes and relationships, reducing its effectiveness in complex scenes. Our model addresses this by leveraging higher intra-modal similarity through integrated visual prompts, enriching cross-modal cosine similarity and supplementing semantic information beyond text prompts alone.

2.2. Open-Vocabulary Semantic Segmentation

Open-vocabulary semantic segmentation (OVSS) enables models to segment arbitrary object categories, including unseen classes. While early zero-shot approaches relied on attribute-based classifiers [22] and word embeddings [63], vision-language models like CLIP [37] and ALIGN [18] have significantly advanced the field through image-text alignment in shared embedding spaces. OVSS methods generally follow either ❶ a two-stage pipeline [8, 9, 12, 17, 23, 55] or ❷ an end-to-end network [53, 54, 56, 69]. Among two-stage approaches, SCAN [27] calibrates CLIP with semantic priors, EBSeg [40] employs balanced decoders with semantic consistency, and USE [48] leverages multi-granularity segments. End-to-end methods have progressed from MaskCLIP’s [68] direct CLIP adaptation to SAN’s [56] sophisticated side-adaptor network. Recent advances like CAT-Seg [7] and SED [52] utilize cost volumes with auxiliary backbones and hierarchical feature fusion, respectively. Building upon these works, we address the limitation of detail degradation during cost volume upsampling by proposing a multi-scale approach that combines both visual and text prompts to better utilize feature information across scales.

2.3. Visual Prompting

Prompting generally refers to providing textual input to an AI model to elicit a specific task. Visual prompting extends this concept by using an image as a prompt, where semantically related reference images guide the model in tasks such as segmentation [65]. Visual prompting can be broadly

categorized into prototype-based and feature-matching approaches. Prototype-based methods, such as PFENet [68] and PANet [46], extract average prototypes of embeddings for each semantic category. Feature-matching methods, like CyCTR [61] and HDMNet [35], use pixel-level correlations between reference and target images to boost segmentation performance. In recent segmentation advancements, GFSS [16] uses learned visual prompts with a transformer decoder to extract prototypes for generalized few-shot segmentation. We employ Stable Diffusion [38] for automatic visual prompt generation aligned with text descriptions, enabling CLIP to produce rich semantic cost volumes through dual-prompt integration.

3. Discussion and Motivation

In this section, we conduct experiments to analyze the modality gap [19, 24] between the aligned image embeddings and text embeddings extracted by pre-trained vision-language models, such as CLIP, as described in Sec. 3.1. Additionally, we explore cost volume visualization to highlight the benefits of visual prompts by comparing the cost volumes generated from text and visual prompts, as shown in Sec. 3.2. Based on the above analysis and observations, we propose combining visual and text prompts to enhance segmentation accuracy, particularly in scenarios involving both seen and unseen categories.

3.1. Modality Gap of the Aligned Image and Text Embeddings

To demonstrate the modality gap [19, 24] between aligned image and text embeddings, we conduct the experiment by following these steps: ❶ Given an image I and its corresponding text prompt P_t , we use the pre-trained CLIP model to extract their embeddings E and T , respectively. ❷ Using the pre-trained Stable Diffusion model [38] to generate the synthetic visual prompt P_v by inputting P_t and then extract its corresponding embedding V using the same CLIP image encoder. ❸ Calculating the cosine similarity between E and T , as well as between E and V , across more than one hundred samples. As shown in Fig. 2, the similarity between E and V is significantly higher than that between E and T , demonstrating that the visual prompt P_v , which shares the same modality as I , can significantly alleviate the embedding modality gap.

3.2. Effectiveness of the Visual Prompt in Semantic Segmentation

To analyze the effectiveness of the visual prompt in semantic segmentation, we conduct the experiment as follows: ❶ Using the pre-trained CLIP image encoder, we extract the image feature from the layer before pooling. ❷ Using the same approach as in Sec. 3.1, we extract the visual prompt embedding V and the text prompt embedding T . ❸ We then

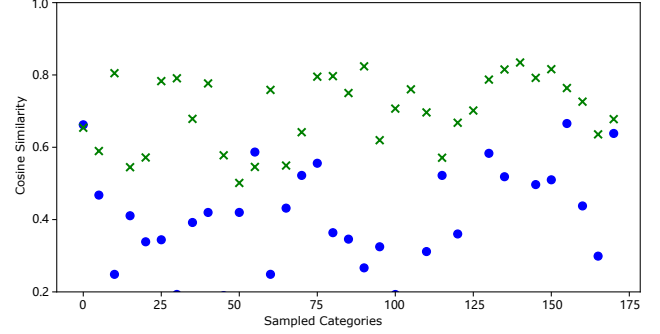


Figure 2. Visualization of cosine similarities comparing image embeddings E with text prompt embeddings T (blue dots) and visual prompt embeddings V (green crosses) across sampled categories.

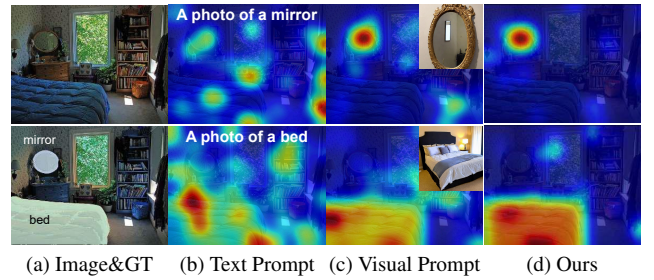


Figure 3. Visualization of cost volume: (a) image with ‘mirror’ and ‘bed’ segments; (b) cost volume with text prompts; (c) cost volume with visual prompts; (d) aggregated cost volume. The top row represents the unseen class ‘mirror,’ and the bottom row represents the seen class ‘bed’.

calculate the similarity between the image feature of each pixel location and the visual prompt embedding V as well as the text prompt embedding T . As shown in the heatmap in Fig. 3, visual prompts yield sharper and more accurate semantic contours than text prompts, particularly for details like “mirror” and “bed” in Fig. 3 (b) and (c).

Inspired by this observation, in this work we propose to leverage both visual and text prompts to enhance the segmentation performance. Specifically, we average the visual and text prompt embeddings and calculate the cosine similarity as described above. As illustrated in Fig. 3 (d), using both visual and text prompts reduces noise and captures more accurate semantic information.

4. Method

Building on insights from Sec. 3, we propose our novel framework **DPSeg** for open-vocabulary semantic segmentation, comprising three core components: a dual-prompt cost volume generation module (see Sec. 4.1), a cost volume-guided decoder (see Sec. 4.2), and a semantic-guided prompt refinement module (see Sec. 4.3), as shown in Fig. 4 and Fig. 5. The dual-prompt cost volume generation module utilizes both text and visual prompts to generate a pixel-level cost volume $\mathcal{F}_c$ with image features. The

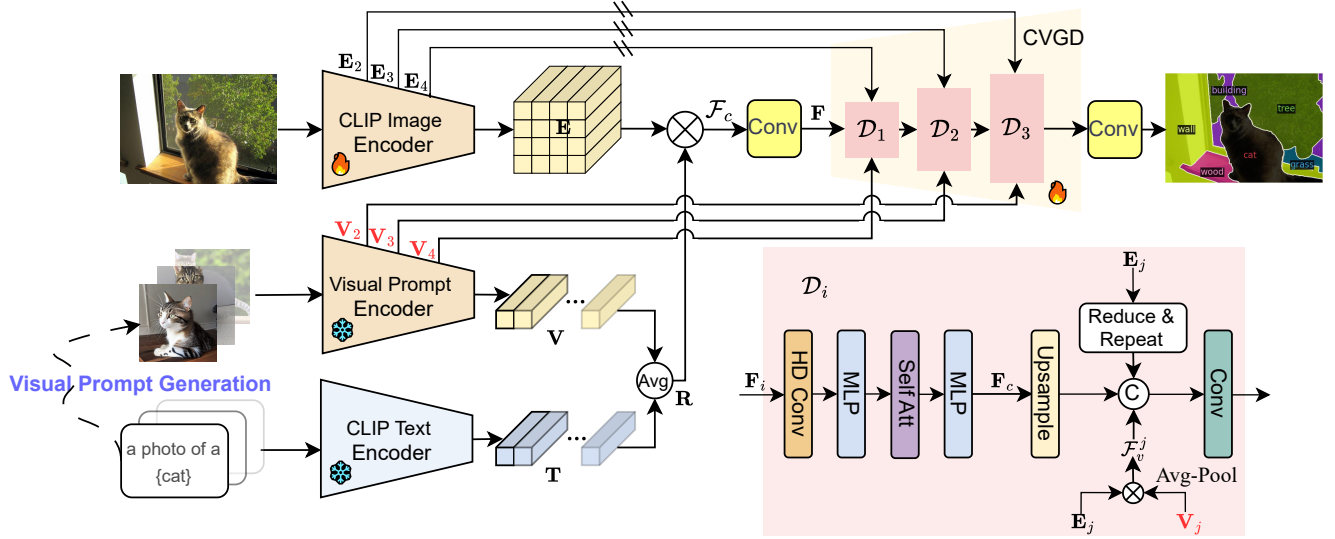


Figure 4. Architecture of our DPSEg network. We begin by generating visual prompts based on text prompt templates, which are then combined with text prompts to create the dual-prompt cost volume. Subsequently, we incorporate multi-scale image features and corresponding visual prompt embeddings into the cost volume-guided decoder (CVGD) in a progressive manner.

cost volume-guided decoder integrates multi-scale cost volumes $\mathcal{F}_v^2, \mathcal{F}_v^3, \mathcal{F}_v^4$ from the image and visual prompts to guide segmentation map prediction. To address misalignment and uncertainty in visual prompt generation, we propose a two-pass inference strategy, semantic-guided prompt refinement, to enhance alignment and improve segmentation accuracy.

4.1. Dual-Prompt Cost Volume Generation

Prior open-vocabulary semantic segmentation approaches [7, 52] mainly rely on text-image feature alignment using CLIP to generate pixel-level cost volumes. However, such cross-modal alignment can be limited in precision due to inherent differences between text and image representations, even with advanced pre-training as discussed in Sec. 3. To address this, we introduce a cross-modal text-visual prompt cost volume generation, which enhances the alignment within the visual domain by introducing higher intra-modality similarity.

Image encoder. For an input RGB image $\mathbf{I}$, we employ a hierarchical encoder to extract image features $\mathbf{E}_j, j \in (2, 3, 4, 5)$ at progressively coarser resolutions, corresponding to scales of $4\times, 8\times, 16\times$, and $32\times$ smaller than the original input size. The final layer, $\mathbf{E}_5$, is processed through an MLP layer to align the dimensionality with the prompt embeddings, yielding the final image embedding $\mathbf{E} \in \mathbb{R}^{H \times W \times D_z}$, where H, W , and D_z represent the height, width, and number of channels respectively.

Text prompt embedding generation. For generating text prompt embeddings, we adopt a diverse set of descriptive templates following the strategies in [7, 13, 23, 52]. These templates provide varied descriptions for each cat-

egory C_k , such as “a photo of a $\{C_k\}$ ”, “a $\{C_k\}$ in the scene”, etc. Each description $P_t^{(k,m)}$ is processed through CLIP’s text encoder, producing text embeddings $\mathbf{T} = \{\mathbf{T}_{k,m} \mid k = 1, \dots, K; m = 1, \dots, M\} \in \mathbb{R}^{K \times M \times D_z}$, where K is the total number of categories, M is the number of templates per category, and D_z is the dimension of the text prompt embeddings, which matches that of the image embeddings for consistency. This variety of templates enhances the semantic richness of each category representation, supporting robust category detection across diverse contexts.

Visual prompt embedding generation. For visual prompt generation, we employ a pre-trained Stable Diffusion model [38] to generate visual prompts by inputting text prompt $\mathcal{P}_t^{(k,m)}$, with each template producing a distinct prompt prototype for the category. The generated visual prompts are processed through a visual prompt encoder, structurally identical to the image encoder. The visual prompt encoder produces an embedding $\mathbf{V} \in \mathbb{R}^{K \times M \times D_z}$, aligned with the text prompt embeddings for seamless integration. Additionally, we extract multi-scale features $\mathbf{V}_j \in \mathbb{R}^{H_j \times W_j \times K \times M \times D_j}$ (for $j = 2, 3, 4$), where H_j and W_j denote the spatial dimensions downsampled by 4, 8 and 16, respectively; K and M represent the category and prototypes of visual prompts, and D_j is the feature dimension at each level j . These multi-scale features further enrich the cost volume, providing fine-grained semantic information that enhances decoding in the segmentation process.

Combined Cost Volume Generation. To utilize both visual and text prompt embeddings, we average them to create a unified representation $\mathbf{R}$, denoted as

$$\mathbf{R} = \text{Avg}(\mathbf{V} + \mathbf{T}).$$

Then we calculate the dual-prompt cost volume $\mathcal{F}_c \in \mathbb{R}^{H \times W \times K \times M}$ using the $\mathbf{R}$ and image feature $\mathbf{E}$, as follows:

$$\mathcal{F}_c(x, y, k, m) = \frac{\mathbf{E}(x, y) \cdot \mathbf{R}(k, m)}{\|\mathbf{E}(x, y)\| \|\mathbf{R}(k, m)\|}, \quad (1)$$

where (x, y) represents the spatial location, and (k, m) denotes the category and template index, respectively. To facilitate efficient processing of this high-dimensional cost volume, we apply a convolution layer independently to each cost slice, producing an initial cost volume embedding. This embedding is then fed into the decoder as $\mathbf{F} \in \mathbb{R}^{(H \times W) \times K \times d_F}$, where d_F is the cost volume embedding dimension.

4.2. Cost Volume-Guided Decoder

Building on the findings in Sec. 3, we design a Cost Volume-Guided Decoder that leverages dual-prompt cost volume and hierarchical encoder structure. At each decoder stage $\mathcal{D}_i$ ($i \in (1, 2, 3)$), the input feature map $\mathbf{F}_i$ is processed through a hybrid dilated convolution (HD Conv) module with a 3×3 kernel and dilation rates of 1, 2, and 4, providing large receptive fields (3×3 , 5×5 , and 9×9) with fewer parameters than standard large-kernel convolutions. These features are merged via element-wise addition, followed by normalization, an MLP with GeLU activation, and a self-attention layer for category-level aggregation.

To integrate multi-scale image features from the encoder, which captures rich local details, we first upsample the feature map $\mathbf{F}_c$ following the self-attention layer by a factor of 2 using deconvolution. Each scale feature $\mathbf{E}_j$, $j \in (2, 3, 4)$ is then reduced by a factor of 16 through convolution, repeated the number of categories N times to align with the category dimension of the upsampled feature map, and finally concatenated with it. As suggested in [7], we avoid back-propagating these intermediate features directly to the image encoder to prevent potential performance degradation.

Instead of upsampling initial cost volume, we employ intra-modal similarity by integrating intermediate-scale image features $\mathbf{E}_j \in \mathbb{R}^{K \times M \times D_j}$, $j \in (2, 3, 4)$, with corresponding multi-scale visual prompt features $\mathbf{V}_j \in \mathbb{R}^{H_j \times W_j \times K \times M \times D_j}$, $j \in (2, 3, 4)$ to compute the visual cost volumes, thereby enhancing semantic coherence across scales. This method captures fine-grained semantic information at multiple levels, yielding visual cost volumes $\mathcal{F}_v^j$ at each stage, calculated as:

$$\mathcal{F}_v^j(x, y, k, m) = \frac{\text{Pool}(\mathbf{V}_j(k, m)) \cdot \mathbf{E}_j(x, y)}{\|\text{Pool}(\mathbf{V}_j(k, m))\| \|\mathbf{E}_j(x, y)\|}. \quad (2)$$

Here, global average pooling is applied along the H_j and W_j . Each channel in the visual cost volume directly aligns with the corresponding decoding feature channels at each

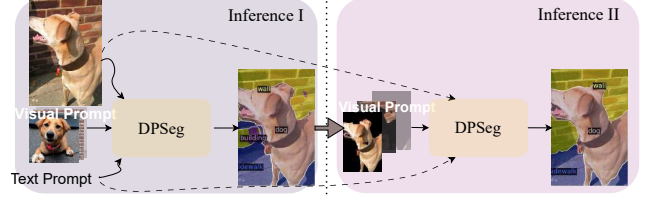


Figure 5. Structure of Semantic-Guided Prompt Refinement.

decoder stage, eliminating the need for additional upsampling and preserving both semantic and spatial details. The final high-dimensional feature map is then mapped to the segmentation output through a convolution layer, enabling accurate predictions across all semantic categories for open-vocabulary semantic segmentation.

4.3. Semantic-Guided Prompt Refinement

The visual prompt generation process during training and initial inference introduces variability, as visual prompts generated from text prompts may not fully align with the input image $\mathbf{I}$. This misalignment can impact segmentation accuracy, particularly for fine-grained details or objects that are not well-represented by the initial visual prompt. To address this challenge, we propose a semantic-guided prompt refinement inference strategy, illustrated in Fig. 5. The first inference (Inference I) pass employs text prompts and initial visual prompts to produce preliminary segmentation results. For each category detected in the first pass, its corresponding segmentation mask is used to crop the objects, which will replace all visual prompts of that category in the second inference pass (Inference II) while maintaining initial prompts for undetected categories and text prompts. This instance-aware refinement enhances the cost volume’s semantic alignment with actual objects in the current image, achieving more precise object boundaries and local details.

5. Experiments

5.1. Setups

Datasets. To acquire object information and evaluate our models, we follow a dataset setup similar to that in [52]. We train our model on the **COCO-Stuff** [3] dataset and evaluate its performance across five additional benchmarks: **Pascal VOC** [11], **Pascal Context** [31] (referred to as *PC-59* and *PC-459*), **ADE20K-150** (referred to as *A-150*) [67], and **ADE20K-847** (referred to as *A-847*) [67]. **COCO-Stuff** is a large-scale semantic segmentation dataset comprising 171 densely annotated classes, with 118k training images, 5k validation images, and 41k test images. **Pascal VOC**, a traditional dataset for segmentation and object detection, includes 20 classes, with 1464 training and 1449 validation images. **Pascal Context** is an extension of **Pascal VOC**, with 1.5k training images and 1.5k validation images, offer-

Method	VLM	Backbone	Training Dataset	A-847	PC-459	A-150	PC-59	PAS-20
ZS3Net [2]	-	ResNet-101	PASCAL VOC	-	-	-	19.4	38.3
LSeg [51]	ViT-B/32	ResNet-101	PASCAL VOC-15	-	-	-	-	47.4
LSeg+ [12]	ALIGN RN-101	ResNet-101	COCO-Stuff	2.5	5.2	13.0	36.0	-
Han et al. [14]	ViT-B/16	ResNet-101	COCO Panoptic [20]	3.5	7.1	18.8	45.2	83.2
ZegFormer [8]	ViT-B/16	ResNet-101	COCO-Stuff	5.6	10.4	18.0	45.5	89.5
SimSeg [55]	ViT-B/16	ResNet-101	COCO-Stuff	7.0	-	20.5	47.7	88.4
OpenSeg [12]	ALIGN	ResNet-101	COCO Panoptic [20] + Loc. Narr. [36]	4.4	7.9	17.5	40.1	-
PACL [33]	ViT-B/16	-	GCC [41] + YFCC[45]	-	-	31.4	50.1	72.3
OVSeg [23]	ViT-B/16	ResNet-101c	COCO-Stuff+COCO Caption	7.1	11.0	24.8	53.3	92.6
CAT-Seg [7]	ViT-B/16	ResNet-101	COCO-Stuff	8.4	16.6	27.2	57.5	93.7
SAN [56]	ViT-B/16	-	COCO-Stuff	10.1	12.6	27.5	53.8	94.0
EBSeg [40]	ViT-B/16	-	COCO-Stuff	11.1	17.3	30.0	56.7	94.6
SCAN [27]	ViT-B/16	Swin-B	COCO-Stuff	10.8	13.2	30.8	58.4	97.0
SED [52]	ConvNeXt-B	-	COCO-Stuff	11.4	18.6	31.6	57.3	94.4
DPSeg (Inference I)	ConvNeXt-B	-	COCO-Stuff	12.0	19.5	32.9	58.1	96.0
DPSeg (Inference II, Ours)	ConvNeXt-B	-	COCO-Stuff	12.5	20.1	33.3	58.4	96.9
LSeg [51]	ViT-B/32	ViT-L/16	PASCAL VOC-15	-	-	-	-	52.3
OpenSeg [12]	ALIGN	Eff-B7 [44]	COCO Panoptic [20] + L.Oc. Narr. [36]	8.1	11.5	26.4	44.8	-
OVSeg [23]	ViT-L/14	Swin-B	COCO-Stuff+COCO Caption	9.0	12.4	29.6	55.7	94.5
Ding et al.	ViT-L/14	-	COCO Panoptic [20]	8.2	10.0	23.7	45.9	-
ODISE [54]	ViT-L/14	-	COCO Panoptic [20]	11.1	14.5	29.9	57.3	-
HIPIE [49]	BERT-B	ViT-H	COCO Panoptic [20]	-	-	29.0	59.3	-
SAN [56]	ViT-L/14	-	COCO-Stuff	13.7	17.1	33.3	60.2	95.5
CAT-Seg [7]	ViT-L/14	Swin-B	COCO-Stuff	10.8	20.4	31.5	62.0	96.6
FC-CLIP [59]	ConvNeXt-L	-	COCO Panoptic [20]	14.8	18.2	34.1	58.4	95.4
EBSeg [40]	ViT-L/14	-	COCO-Stuff	13.7	21.0	32.8	60.2	96.4
SCAN [26]	ViT-L/14	Swin-B	COCO-Stuff	14.0	16.7	33.5	59.3	97.2
USE+SAM [47]	ViT-L/14	Swin-B	COCO-Stuff	13.3	14.7	37.0	57.8	-
SED [52]	ConvNeXt-L	-	COCO-Stuff	13.9	22.6	35.2	60.6	96.1
DPSeg (Inference I)	ConvNeXt-L	-	COCO-Stuff	14.9	23.5	36.4	62.0	97.4
DPSeg (Inference II, Ours)	ConvNeXt-L	-	COCO-Stuff	15.7	24.1	37.1	62.3	98.5

Table 1. Comparison with state-of-the-art methods on five open-vocabulary semantic segmentation test sets. mIoU results are shown, with the best in bold and the second-best underlined. In both configurations of the VLM model, our DPSeg under Inference I and II achieves superior performance, with Inference II consistently outperforming all other SOTA methods.

ing two types of annotations: *PC-59* (59 classes) and *PC-459* (459 classes). The **ADE20K** dataset consists of 20k training images and 2k validation images and is split into two test sets: *A-150* with 150 annotated classes, and *A-847* with 847 common semantic categories.

Metrics. We evaluate our model using Intersection over Union (IoU). Our model is evaluated without resorting to finetuning in various settings, demonstrating the enhancements it achieves.

5.2. Implementation Details

Network Architectures. We employ the OpenAI pre-trained CLIP model [37] as the backbone, using either ConvNeXt-B or ConvNeXt-L as the image encoder and visual prompt encoder. For the text encoder, the text embedding dimension D_z is set to 640 for ConvNeXt-B and 768 for ConvNeXt-L. The initial cost volume fed into the decoder is set to a dimension of 128, with the output hidden dimensions of each decoder stage as [62, 32, 16]. For consistency, visual prompts are generated at 768×768 resolution, matching the input image size during training and inference. To improve training efficiency, the text and visual prompt encoders are frozen, with only the image encoder and cost volume-guided decoder being trained. Training uses a per-pixel binary cross-entropy loss. Our models are implemented in PyTorch [34] and Detectron2 [50]. We employ the AdamW optimizer with an initial learning rate of

2×10^{-4} , weight decay of 1×10^{-4} , and an additional factor $\alpha = 0.01$ to mitigate overfitting. The mini-batch size is set to 4, and models are trained for 80k iterations on two V100 GPUs.

5.3. Quantitative Results

As shown in Table 1, in the base-VLM configuration, our approach achieves significant gains over state-of-the-art methods like SED [52], with improvements of +1.1%, +1.5%, +2.5%, and +1.1% mIoU on the *A-187*, *PC-459*, *A-150*, and *PC-59* datasets, respectively. Between our two inference stages, Inference II shows an additional gain of +0.87% mIoU over Inference I. In the *Large-scale* VLM configuration, our method further outperforms SED, achieving gains of +1.8%, +1.5%, +1.9%, +1.7%, and +2.4% mIoU across all datasets, with Inference II yielding an extra +0.92% mIoU over Inference I. These results underscore the effectiveness of our proposed method in advancing OVSS.

5.4. Qualitative Results

Fig. 6 shows segmentation examples from our model. In the first four columns, we display samples from the *ADE-150* and *PC-59* datasets alongside the ground truth, illustrating our model’s strong performance across diverse scenarios. Our segmentation results not only capture large foreground objects in both seen and unseen classes but also de-



Figure 6. Visualization examples of our model on open vocabulary semantic segmentation. **First four columns:** We present results from our model alongside the corresponding ground truth. **Fifth and sixth columns:** SED [52] results are shown for comparison, highlighting our model’s improved segmentation of small foreground objects, such as the mouse, bottle, and tennis racket.

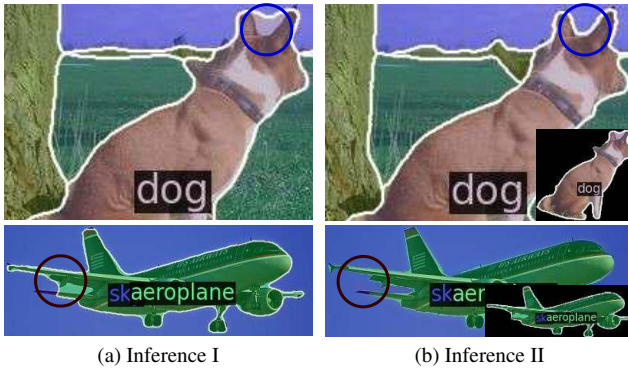


Figure 7. Comparison between inference I and II with our refinement strategy.

test additional categories not annotated in the ground truth, such as “wood,” “fence,” and “chair” in the first, second, and fourth images of the first row. We also compare three cases between SED [52] and our model. As shown in the last two columns of Fig. 6, our method accurately segments smaller objects near larger ones, such as the mouse, bottle, and tennis racket, effectively preserving semantic information across scales. In Fig. 7, we compare two cases of our model between Inference I and Inference II. The results demonstrate our method’s effectiveness in enhancing semantic alignment and capturing finer segmentation details

based on the initial segmentation output.

5.5. Ablation Study

For simplicity, all ablation studies are conducted using the strategy and results from Inference I.

Prompt strategy and cost volume generation. In this section, we conduct ablation studies to explore how different prompt strategies and cost volume generation strategies affect the segmentation results. From Table 2 (a), we observe that the visual-only prompt outperforms the text-only prompt. By averaging the visual and text embedding, achieving better overall performance. In Table 2 (b), we explore two additional cost volume generation methods: ① generating separate cost volumes for text and visual prompts, then concatenating them, and ② directly averaging the two cost volumes. The results clearly show that our approach—averaging the embeddings of both modalities before calculating cosine similarity—yields superior performance.

Cost volume-guided decoder. We examine the effect of various cost volume guidance strategies in our decoder, as shown in Table 3. When using only the cost volume $\mathcal{F}_c$ generated by text and visual embeddings, segmentation performance is lowest (first row). Performance steadily improves as we progressively integrate multi-scale visual cost vol-

	Prompt Strategy	A-847	PC-459	A-150	PC-59	PAS-20
(a)	T	10.4	17.4	30.6	56.2	93.4
	V	11.1	18.0	31.2	56.9	94.5
	Avg(T , V) (Ours)	12.0	19.5	32.9	58.1	96.0
(b)	Cost Volume Generation	A-847	PC-459	A-150	PC-59	PAS-20
	Cat(cos(T , E), cos(V , E))	11.7	19.2	32.3	57.8	95.4
	Avg(cos(T , E), cos(V , E))	11.9	19.4	32.7	58.0	95.7
	$\mathcal{F}_c$ (Ours)	12.0	19.5	32.9	58.1	96.0

Table 2. **Ablation study on prompt types and cost volume generation strategies.** Results are shown for text-only (**T**), visual-only (**V**), and combined text-visual prompt embeddings, along with comparisons of three cost volume generation solutions

Fusion Strategy	A-847	PC-459	A-150	PC-59	PAS-20
$\mathcal{F}_c$	10.6	17.9	31.6	57.0	94.9
$\mathcal{F}_c, \mathcal{F}_v^2$	11.4	18.8	32.1	57.6	95.5
$\mathcal{F}_c, \mathcal{F}_v^{2,3}$	11.7	19.2	32.5	57.9	95.7
$\mathcal{F}_c, \mathcal{F}_v^{2,3,4}$ (Ours)	12.0	19.5	32.9	58.1	96.0
$\mathcal{F}_c, \mathcal{F}_c(2\times, 4\times, 8\times)$	11.4	19.0	32.4	58.0	95.6

Table 3. **Ablation study on cost volume guidance strategies.** $\mathcal{F}_c$ denotes the cost volume in our model, while $\mathcal{F}_v^i$ represents cost volume at layer i , calculated from different levels of the image features and visual prompts. $\mathcal{F}_c(2\times)$ refers to the method where the cost volume $\mathcal{F}_c$ is upsampled and concatenated.

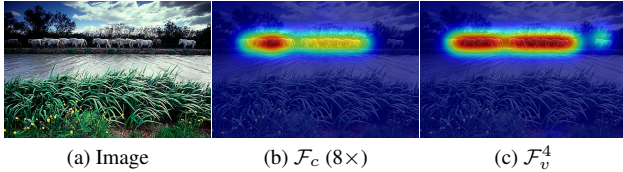


Figure 8. **Comparison of the visualized cost volumes.** Evaluating the effects of cost volume by comparing the (b) $8\times$ upsampled cost volume with the (c) the cost volume derived from 4th intermediate feature and visual prompt.

umes $\mathcal{F}_v^{2,3,4}$, achieving an average increase of 1.8% mIoU across all datasets when all scales are incorporated (second and third rows). Replacing $\mathcal{F}_v^{2,3,4}$ with an upsampled $\mathcal{F}_c$ results in a performance drop (last row). Fig. 8 illustrates how our multi-scale cost volume captures finer semantic details, such as a single horse standing apart from a crowd—details that upsampled cost volumes fail to capture. This progressive integration of multi-scale cost volumes preserves spatial details, enhances multi-scale feature utilization, and maintains semantic consistency across scales, enabling our model to capture fine details for improving segmentation accuracy.

Number of Templates. Table 4 shows the results of an ablation study examining the impact of the number of templates on segmentation performance. In the experiment, each text template is paired with a corresponding visual

Number of Templates	A-847	PC-459	A-150	PC-59	PAS-20
1	3.6	7.8	15.6	32.1	72.3
10	6.9	11.2	20.3	41.3	81.7
20	7.8	14.9	27.9	47.3	86.7
40	10.1	16.8	29.6	55.9	91.8
80	12.0	19.5	32.9	58.1	96.0

Table 4. **Ablation study on the number of prompt templates.** Both text and visual prompts are used.

Visual Prompt Generation	A-847	PC-459	A-150	PC-59	PAS-20
(1) 20% Gaussian Noise	12.0	19.2	32.5	57.9	95.5
(2) 60% Gaussian Noise	11.6	18.8	32.9	57.6	95.0
(3) 80% Gaussian Noise	11.5	18.7	31.7	57.3	94.7
(4) LayerDiffuse[62]	12.1	19.4	32.7	58.0	95.8
(5) SD-V1.5	12.0	19.5	32.9	58.1	96.0
(6) SD-V3[10]	12.2	19.6	32.8	58.3	96.3
(7) SED (text prompt only)	11.4	18.6	31.6	57.3	94.4

Table 5. **Ablation study on visual prompts generation.** Methods(1)–(3) use SD-v1.5 with varying Gaussian noise levels (e.g., 80% means noise variance is 80% of the maximum), methods(4)–(6) employ LayerDiffuse, SD-v1.5, and SD-v3, respectively, and method(7) shows SED segmentation performance.

prompt prototype. As the number of templates increases, we observe a significant improvement in performance. This result suggests that a diverse set of templates better captures variations in real-world scenarios, as a single prompt may fail to represent all contexts. For example, ‘a photo of a bird’ may not accurately depict an image with multiple birds, which would be better represented by ‘a photo of many birds’.

Quality of visual prompts. The quality of visual prompts generated by different generation models varies, as shown in Table 5. Our experimental results highlight the robustness of our approach: even when using low-quality visual prompts (e.g., 80% Gaussian noise), our method consistently outperforms the text-prompt-only baseline (e.g., SED) in segmentation. Furthermore, employing different text-to-image generation models ((4)–(6) in Table 5) leads to minimal performance variation. These findings reinforce our commitment to integrating advanced generation techniques to enhance OVSS for both research and real-world applications.

6. Conclusion

In conclusion, we introduce **DPSeg**, a novel dual-prompt framework for open-vocabulary semantic segmentation (OVSS). Our approach synergistically combines text and visual prompts with multi-scale features, while introducing an innovative refinement strategy to enhance semantic alignment and fine-grained detail preservation. Extensive benchmarks validate its state-of-the-art performance, demonstrating the effectiveness of our dual-prompt cost volume learning approach.

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